



ESMU MODBUS通信协议/ESMU MODBUS COMMUNICATION PROTOCOL (TCP)

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修订

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A0	储能技术 部	2018-12-7	建立			
A1	储能技术 部	2019-04-09	增加 2 台空调湿度			
A2	储 能 技 术部	2019-05-30	增加与 PCS、BMS 通信告警			
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V1	宋瑞祥	2020-12-14	增加单体电压至 700 串			
V1.1	宋瑞祥	2021-3-4	修改寄存器内容和寄存器地址（5.1.X）			
V1.2	宋瑞祥	2021-4-2	修改 BMS 堆当前状态、BMS 簇当前状态、BMS 充放电状态的文字定义			
V1.3	宋瑞祥	2021-6-17	5.1.1 中增加 8 个 IO 扩展盒 DI 状态			
V1.4	宋瑞祥	2021-7-21	1、增加模块过压报警、端子过温报警和采集数据、堆离散中烟感水浸门禁故障点去掉（后面地址上移）、增加温度、电压采集故障 2、5.1.2 保持寄存器中地址 504~523 修改为簇 N 维护控制			
V1.5	宋瑞祥	2021-11-11	1、增加累计充电时间、累计放电时间 2、将 RTC 对时功能修改到保持寄存器中（0x03/0x06/0x10 功能码）			
V1.4.1	钱程隆	2023-1-3	在 V1.4 基础上修改 1、中英文 2、修改通讯机制描述			
V1.6	宋瑞祥	2023-2-9	在 V1.4.1 基础上修改 1、增加累计充电时间、累计放电时间 2、将 RTC 对时功能修改到保持寄存器中（0x03/0x06/0x10 功能码） 3、离散输入寄存器 62-69 改成预留 4、保持寄存器 530 为心跳			

			5、保持寄存器 1000-2000 定义为项目外设信息 6、增加状态补充描述，增加下电中和下电完成两个状态			
V2.1	宋瑞祥	2023-6-30	5.1.3 中删除允许最大放电电流、允许最大充电电流的偏移量			



目录

目录	I
1. 概述/SUMMARY	1
1.1. 参考文献/REFERENCES	1
1.2. 缩略语/ABBREVIATIONS	2
2. 通信方式/COMMUNICATION MODE	2
2.1. 物理层接口/INTERFACE	2
2.2. 通信机制/COMMUNICATION MECHANISM.....	2
2.2.1. 标准模式/Standard Mode	2
3. 信息类型及信息结构/INFORMATION TYPE AND INFORMATION STRUCTURE	3
3.1. 数据类型/DATA TYPE	3
3.2. 命令结构/COMMAND STRUCTURE.....	4
3.3. 数据存储和传输顺序/DATA STORAGE AND TRANSFER ORDER	5
3.3.1. 16bit 数据类型字节存储及传输顺序/16-bit data storage and transfer order	5
3.3.2. 32bit 数据类型字存储及传输顺序/32-bit data storage and transfer order	6
3.3.3. 字符串数据类型存储及传输顺序/ String data storage and transfer order	6
3.3.4. 布尔型数据类型存储及传输顺序/BOOL data storage and transfer order	7
3.4. 对象类型/OBJECT TYPE	7
3.4.1. ESMU 设备/ESMU	7
3.4.2. ESBCM.....	8
4. 命令解析/COMMAND	9
4.1. 遥测命令/TELEMETRY COMMAND	9
4.1.1. 读保持寄存器 03H/Read Holding Register 03H	9
4.1.2. 读输入寄存器 04H/Read input register 04H	10
4.2. 遥信命令/ REMOTE SIGNALING COMMAND.....	11
4.3.1 读离散输入寄存器 02H/ Read discrete input register.....	11
4.3. 遥调命令/ REMOTE ADJUSTMENT COMMAND	14
4.3.1. 校时 41H/Timing 41H.....	14
4.3.2. 单项调节保持寄存器 06H/ Single regulation holding register 06H	15
4.3.3. 区块调节保持寄存器 10H/ Block adjustment holding register 10H	16
4.4. 异常应答机制/ EXCEPTION RESPONSE MECHANISM	18
4.5. 常用事务参考定义/ COMMON TRANSACTION REFERENCE DEFINITIONS.....	21
4.5.1. ESMU 设备对象常见事务/ Common transactions of ESMU device objects	21
4.5.2. ESBCM 对象常见事务/ Common transactions of ESBCM device objects.....	21
5. 寄存器分布 /REGISTER DISTRIBUTION	22
5.1. ESMU 设备对象寄存器分布/ESMU REGISITER DISTRIBUTION	22
5.1.1. 离散输入寄存器表/ Discrete Input Register Table	22
5.1.2. 保持寄存器表/ Holding register table	35
5.1.3. 输入寄存器表/ Input register table	37

1. 概述/Summary

本规约用于规范管理储能电池管理系统电池堆管理单元 ESMU 与后台监控/EMS 之间的通信。协议参考了 MODBUS 标准在 TCP/IP 上的实现，根据储能系统的系统特点，对通信机制和通信管理进行了一定的改进，同时支持标准模式。

This protocol is used to regulate and manage the communication between the battery stack management unit ESMU of the energy storage battery management system and the background monitoring/EMS. The protocol refers to the implementation of MODBUS standard on TCP/IP, and improves the communication mechanism and communication management according to the system characteristics of the energy storage system, while supporting the standard mode.

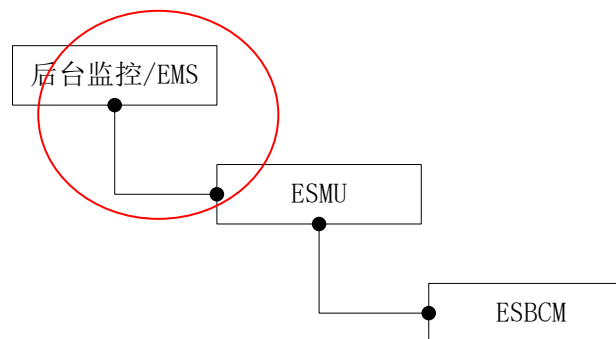


图 1.1 储能系统通信层次

1.1. 参考文献/References

GB/T 19582-1 2008 《基于 Modbus 协议的工业自动化网络规范 第 1 部分：Modbus 应用协议》

GB/T 19582-3 2008 《基于 Modbus 协议的工业自动化网络规范 第 3 部分：Modbus 协议在 TCP/IP 上的实现指南》

GB/T 19582-1 2008 Industrial Automation Network Specification Based on Modbus Protocol Part 1: Modbus Application Protocol

GB/T 19582-3 2008 Industrial Automation Network Specification Based on Modbus Protocol Part 3: Implementation Guide of Modbus Protocol over TCP/IP



1.2. 缩略语/Abbreviations

ESMU: 电池管理系统电池堆管理单元/ Battery management system battery System management unit

ESBCM: 电池管理系统电池簇管理单元/ Battery management system battery String management unit

EMS: 能量管理系统/ Energy Management System

2. 通信方式/Communication Mode

2.1. 物理层接口/Interface

ESMU 通过 RJ45 接口接入以太网/ ESMU accesses Ethernet through RJ45 interface。

2.2. 通信机制/Communication mechanism

本规约定义 ESMU 与后台监控/EMS 之间为双向通信，通信模式：“标准模式”。

The protocol defines two-way communication between ESMU and background monitoring/EMS, and the communication mode is "standard mode".

2.2.1. 标准模式/Standard Mode

ESMU 作 TCP 服务器，默认端口 502。后台监控作客户端，端口大于 1024。只能客户端发起问询。ESMU 应该能够修改服务端口，后台监控端也应该提供远程端口配置功能，以应对 ESMU 端口被占用的情况。正常情况，任何一方发送命令，另一方都会应答。

在该模式下，后台监控主动连接 ESMU。

ESMU 如果 30 秒没有收到后台的命令，断开与后台的连接。

ESMU is the TCP server, and the default port is 502. The background monitoring is used as the client, and the port is greater than 1024. Only the client can initiate queries. The ESMU should be able to modify the service port, and the background monitoring terminal should also provide the remote port configuration function to deal with the ESMU port occupation. Normally, if either party sends a command, the other party will respond.

In this mode, background monitoring actively connects to ESMU.

If the ESMU does not receive a command from the background within 30 seconds, disconnect it from the background.

3. 信息类型及信息结构/Information Type and Information Structure

3.1. 数据类型/Data Type

表 3-1 数据类型表/ Data Type Table

数据类型	符号	说明
8bit 无符号整数 Unsigned integer	BYTE	0~255
16bit 无符号整型 Unsigned integer	UINT	0~65535
16bit 有符号整型 signed integer	INT	-32768~32767
32bit 浮点型 Float	FLOAT	IEEE 标准 32bit 浮点数
32bit 无符号整型 Unsigned integer	UDINT	
32bit 有符号整型	DINT	

signed integer		
32bit BCD 码	8BCD	8 位压缩 BCD 码, 如十进制 8092101 表示为 08092101H。 An 8-bit compressed BCD code, such as decimal 8092101 is represented as 08092101H
16 字节字符串 16Bytes String	16chString	
32 字节字符串 32Bytes String	32chString	
布尔型	BOOL	只有“0”、“1”两种状态。仅用于离散输入寄存器。 There are only two states: "0" and "1". Used only for discrete input registers.

3.2. 命令结构/Command Structure

表 3-2 命令结构表/Table3-2 Command Structure Table

字段/Field	字节数 /Bytes	描述/description
事务处理标识符 The transaction identifier	2	UINT 型数据, 发送/应答事务处理识别。发送和应答命令中的事务处理标识符一致。范围 0~65535。参考事务定义见 4.7 UINT-type data, send/reply transaction recognition. The transaction identifiers in the send and reply commands are consistent. Range 0~65535. See 4.7 for the reference transaction definition
协议标识符	2	0

Protocol identifier		
长度/length	2	UINT 型数据，表示随后字节数 UINT-type data, representing the number of subsequent bytes
单元标识符 /Unit identifier function code	1	BYTE 型数据，对象的地址。范围 1~247。 BYTE type data, the address of the object. Range 1~247.
功能码 /Function code	1	BYTE 型数据。见“ 命令解析 ”。正常情况下发送和应答的功能码一致，异常情况下参见“ 异常应答机制 ” BYTE-type data. See Command parsing. Under normal circumstances, the function code of sending and replying is the same, see "Abnormal Response Mechanism" under abnormal conditions
数据/data	N	见“ 命令解析 ”。 Command parsing.

3.3. 数据存储和传输顺序/Data Storage and Transfer Order

3.3.1. 16bit 数据类型字节存储及传输顺序/16-bit data storage and transfer order

对于 16bit 数据类型，先传输高字节，再传输低字节。例如 3A56H，先传送 3AH，再传送 56H。

输入寄存器和保持寄存器都是 16bit 寻址的寄存器。可以认为单个寄存器的位存储顺序也是高字节在前，低子字节在后。

For the 16bit data type, the high byte is transmitted first, and then the low byte is transmitted. For example, 3A56H is transmitted first and then 56H.

The input register and holding register are 16bit addressable registers. It can be considered that the bit storage order of a single register is also high byte first and low sub byte last.

3.3.2. 32bit 数据类型字存储及传输顺序/32-bit data storage and transfer order

对于 32bit 浮点型、32bit 整型等 32bit 数据类型，先传输高 16bit 字，再传输低 16bit 字。比如 8DF377A2H 数据，先传输 8DF3H，再传输 77A2H，字节流为 8DH、F3H、77H、A2H。

在输入寄存器和保持寄存器中，32bit 数据占用连续的两个寄存器地址。相应低地址存储高 16 位，高地址存放低 16 位。8DF377A2H 数值如果用地址 12、13 两个寄存器来存放，则地址 12 寄存器存放 8DF3H，地址 13 寄存器存放 77A2H。

For 32bit data types such as 32bit floating point type and 32bit integer type, the high 16bit words are transmitted first, and then the low 16bit words are transmitted. For example, 8DF377A2H data is transmitted first to 8DF3H, and then to 77A2H. The byte streams are 8DH, F3H, 77H, and A2H.

In the input register and holding register, 32bit data occupies two consecutive register addresses. The corresponding low address stores the high 16 bits and the high address stores the low 16 bits. If the 8DF377A2H value is stored in the address 12 and 13 registers, the address 12 register stores 8DF3H and the address 13 register stores 77A2H.

3.3.3. 字符串数据类型存储及传输顺序/ String data storage and transfer order

字符串数据传输时先传输较靠前的字符。

字符串数据类型在输入寄存器和保持寄存器存储时，顺序上较靠前的字符存储在较低地址的寄存器中。在一个寄存器中，较靠前的 8bit 字符存储在高字节上，较靠后的 8bit 字符存储在低字节上。

When string data is transmitted, the characters that precede are transmitted first.

When the string data type is stored in the input register and the holding register, the characters that are higher in order are stored in the lower address register. In a register, the first 8 bit characters are stored in the high byte, and the last 8 bit characters are stored in the low byte.



3.3.4. 布尔型数据类型存储及传输顺序/BOOL data storage and transfer order

布尔型数据主要分布在离散输入寄存器中。

离散输入寄存器实际以字节为单位存储和传输。较低地址的寄存器存储在一个字节的较低位上。传输时，先传输较低地址的寄存器值。存储较低地址的 8 个寄存器的字节先于存储较高地址的 8 个寄存器的字节传输。

例如：连续的 16 个离散输入寄存器的值从低地址到高地址依次为：1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1。传输时分两个字节传输，先传输二进制 1, 1, 0, 1, 1, 1, 0, 0 八个位，实际存储的值为二进制 00111011（注意低位先传），即 3BH。再传送 1, 1, 0, 1, 1, 1, 0, 1 八个位，实际存储的值为 10111011B，即 BBH。

Boolean data is mainly distributed in discrete input registers.

Discrete input registers are actually stored and transmitted in bytes. The register with the lower address is stored in the lower bit of a byte. When transmitting, the register value of the lower address is transmitted first. The bytes of the 8 registers storing the lower address are transmitted before the bytes of the 8 registers storing the higher address.

For example, the values of 16 consecutive discrete input registers from low address to high address are: 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1. When transmitting, two bytes are transmitted. First, eight bits of binary 1, 1, 0, 1, 1, 1, 0, 0 are transmitted. The actual stored value is binary 00111011 (note that the low bit is transmitted first), that is, 3BH. Then transmit eight bits of 1, 1, 0, 1, 1, 1, 0 and 1, and the actual stored value is 10111011B, that is, BBH.

3.4. 对象类型/Object Type

3.4.1. ESMU 设备/ESMU

该对象包含所接电池组数，设备识别码，系统版本号，设备工作状态，所属站信息等。

The object includes the number of connected battery String, equipment identification code, system version number, equipment working status, station information, etc.

3.4.2. ESBCM

该对象包含了该组蓄电池的基本属性以及各类测量值和状态值，还有所使用的采集设备的相关信息。

This object contains the basic attributes of this Battery Sting, various measured values and status values, as well as relevant information about the collection equipment used.

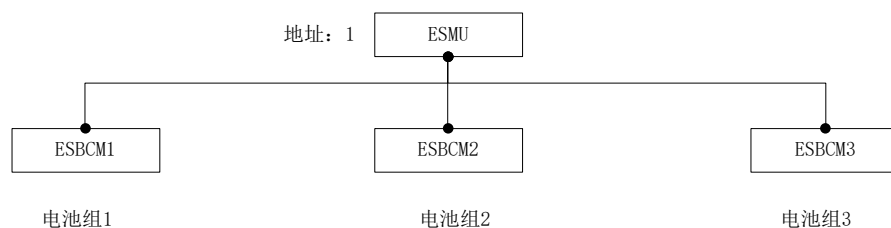


图 3.4.1 对象逻辑结构图/ Object logic structure diagram

图 3.4.1 展示了智能蓄电池监测系统的对象逻辑结构。命令可以直接寻址 ESMU 对象和 ESBCM 对象。ESMU 对象相当于一个 MODBUS 网关，将接收到的发送给 ESBCM 对象的 MODBUS 命令转发给相应的 ESBCM 对象。

ESMU 对象的地址（单元标识符）是 1。各组 ESBCM 对象的地址（单元标识符）为组号加上 1。例如 BMS 系统图如图 3.4.1，需要召唤第 1 组的电池数据，召唤地址为 2；召唤第 2 组的电池数据，召唤地址为 3；召唤第 3 组的电池数据，召唤地址为 4，以此类推。

Figure 3.4.1 shows the object logic structure of the intelligent battery monitoring system. Commands can directly address ESMU objects and ESBCM objects. The ESMU object is equivalent to a MODBUS gateway, which forwards the received MODBUS commands sent to the ESBCM object to the corresponding ESBCM object.

The address (unit identifier) of the ESMU object is 1. The address (unit identifier) of each group of ESBCM objects is the group number plus 1. For example, the BMS system diagram is shown in Figure 3.4.1. The first group of battery data needs to be called, and the calling address is 2; Summon the battery data of the second group, and the calling address is 3; Call the battery data of the third group. The call address is 4, and so on.

4. 命令解析/Command

4.1. 遥测命令/Telemetry command

4.1.1. 读保持寄存器 03H/Read Holding Register 03H

功能码：03H

发起方：监控后台

表 4-2-1-1 读保持寄存器命令发送数据

字段	符号	字节数	数据类型	说明
起始寄存器地址	addr	2×1	UINT	
寄存器个数	cnt	2×1	UINT	<=120

应答：

表 4-2-1-2 读保持寄存器命令应答数据

字段	符号	字节数	数据类型	说明
后续字节数	len	1	BYTE	len = 2×cnt
数据	data	2×cnt	UINT	

Function Code：03H

Request: Monitor the background

Table 4-2-1-1 Read hold register command to send data

Field		Bytes	Data Type	Remark
-------	--	-------	-----------	--------

Start register address	addr	2×1	UINT	
Number of registers	cnt	2×1	UINT	≤ 120

Response:

Table 4-2-1-2 Read hold register command reply data

Field		Bytes	Data Type	Remark
Number of subsequent bytes	len	1	BYTE	$\text{len} = 2 \times \text{cnt}$
Data	data	$2 \times \text{cnt}$	UINT	

4.1.2. 读输入寄存器 04H/Read input register 04H

功能码: 04H

发起方: 监控后台

表 4-2-2-1 读输入寄存器命令发送数据

字段	符号	字节数	数据类型	说明
起始寄存器地址	addr	2×1	UINT	
寄存器个数	cnt	2×1	UINT	≤ 120

应答:

表 4-2-2-2 读输入寄存器命令应答数据

字段	符号	字节数	数据类型	说明
----	----	-----	------	----

后续字节数	len	1	BYTE	len = 2 × cnt
数据	data	2 × cnt	UINT	

Function Code: 04H

Request: Monitor the background

Table 4-2-2-1 Read input register command to send data

Field		Bytes	DataType	Remark
Start register address	addr	2 × 1	UINT	
Number of registers	cnt	2 × 1	UINT	≤120

Response:

Table 4-2-2-2 Read input register command reply data

Field		Bytes	DataType	Remark
Number of subsequent bytes	len	1	BYTE	len = 2 × cnt
Data	data	2 × cnt	UINT	

4.2. 遥信命令/ Remote signaling command

4.3.1 读离散输入寄存器 02H/ Read discrete input register

功能码：02H
发起方：监控后台

表 4-3-1-1 读取离散输入寄存器发送命令

字段	符号	字节数	数据类型	说明
起始寄存器地址	addr	2×1	UINT	
寄存器个数	cnt	2×1	UINT	<=2000

应答：
表 4-3-1-2 读取离散输入寄存器应答命令

字段	符号	字节数	数据类型	说明
后续字节数	len	1	BYTE	当请求寄存器个数为 8 的倍数，len = cnt / 8。否则 len = (cnt / 8) + 1
数据	data	=len	BOOL	

说明：离散输入寄存器表内的数据类型都是 **BOOL** 型，只有“0”，“1”两种状态。地址较小的寄存器分布在一个字节存储区域的较低位。比如读取地址 500 到 507 这 8 个寄存器，传输过程中存放在 1 个字节中，该字节的最低位 **BIT0** 存放地址 500 的寄存器值，最高位 **BIT8** 存放地址 507 的寄存器值。

当请求寄存器个数不是 8 的倍数时，最后一个字节中剩余的位一律用“0”填充。比如读取 500 到 509 这 10 个寄存器，实际这 10 个寄存器的值都为“1”。则不管地址 510 到 515 这 6 个寄存器的值是多少，应答数据都为 **FFH,03H**。注意 PC 在解析应答时不能将填充的剩余位置解析为 510~515 地址的值。

Function Code: 02H
Request: Monitor the background
Table 4-3-1-1 Read discrete input register command to send data

Field		Bytes	Data Type	Remark
Start register	addr	2×1	UINT	

address				
Number of registers	cnt	2×1	UINT	≤ 2000

Response:

Table 4-2-2-2 Read input register command reply data

Field		Bytes	DataType	Remark
Number of subsequent bytes	len	1	BYTE	When the number of request registers is a multiple of 8, $\text{len} = \text{cnt} / 8$. Otherwise $\text{len} = (\text{cnt} / 8) + 1$
Data	data	Len	BOOL	

Note: The data type in the discrete input register table is BOOL, and there are only two states: "0" and "1". Registers with smaller addresses are distributed in the lower bits of a byte storage area. For example, the eight registers reading addresses 500 to 507 are stored in one byte during transmission. The lowest bit of this byte BIT0 stores the register value of address 500, and the highest bit BIT8 stores the register value of address 507.

When the number of request registers is not a multiple of 8, the remaining bits in the last byte are all filled with "0". For example, when reading the 10 registers from 500 to 509, the actual values of the 10 registers are "1". The response data is FFH, 03H, regardless of the values of the six registers from addresses 510 to 515. Note that the PC cannot parse the remaining filled positions into 510~515 address values when parsing the response.

4.3. 遥调命令/ Remote adjustment command

4.3.1. 校时 41H/Timing 41H

功能码/Function code: 41H
发起方/Request site: 监控后台/EMS
对象站址/Target site: ESMU 设备对象地址/ESMU

表 4-4-1-1 校时命令发送数据/ Time calibration command sends data

字段/Field	字节数/Bytes	数据类型/Type	范围/Range
长度/Length	1	BYTE	= 7
年/Year	1	BYTE	0~37
月/Month	1	BYTE	1~12
日/Day	1	BYTE	1~31
时/Hour	1	BYTE	0~23
分/Minute	1	BYTE	1~59
毫秒/Millisecond	2×1	UINT	0~59999

应答/Response: 设置后的系统时间/ System time after setting

表 4-4-1-2 校时命令应答数据/ Time calibration command response data

字段/Field	字节数/Bytes	数据类型/Type	范围/Range
长度/Length	1	BYTE	= 7
年/Year	1	BYTE	0~37
月/Month	1	BYTE	1~12
日/Day	1	BYTE	1~31

时/Hour	1	BYTE	0~23
分/Minute	1	BYTE	1~59
毫秒/Millisecond	2×1	UINT	0~59999

4.3.2. 单项调节保持寄存器 06H/ Single regulation holding register 06H

功能码：06H

发起方：监控后台

表 4-4-2-1 单项调节保持寄存器命令发送数据

字段	符号	字节数	数据类型	说明
寄存器地址	addr	2×1	UINT	
值	val	2×1	UINT	

应答：

表 4-4-2-2 单项调节保持寄存器命令应答数据

字段	符号	字节数	数据类型	说明
寄存器地址	addr	2×1	UINT	与询问的 addr 相同
值	val	2×1	UINT	设置后该寄存器的值

Function Code: 06H

Request: Monitor the background

Table 4-4-2-1 Single regulation holding register command sends data

Field		Bytes	Data Type	Remark
Register address	addr	2 × 1	UINT	
Value	Val	2 × 1	UINT	

Response:

Table 4-4-2-2 Single regulation holding register command response data

Field		Bytes	Data Type	Remark
Register address	addr	2 × 1	UINT	Same as Request
Value	val	2 × 1	UINT	Same as Request

4.3.3. 区块调节保持寄存器 10H/ Block adjustment holding register 10H

发起方：监控后台

功能码：10H

表 4-4-3-1 区块调节保持寄存器命令发送数据

字段	符号	字节数	数据类型	说明
起始寄存器地址	addr	2 × 1	UINT	
寄存器个数	cnt	2 × 1	UINT	≤120

后继数据长度	len	1	BYTE	$len = 2 \times cnt$
值	val	$2 \times cnt$	UINT	

应答:
表 4-4-3-2 区块调节保持寄存器命令应答数据

字段	符号	字节数	数据类型	说明
寄存器地址	addr	2×1	UINT	与询问的 addr 相同
已修改寄存器个数	cnt	2×1	UINT	与询问的 cnt 值相同

Function Code: 10H
Request: Monitor the background

Table 4-4-3-1 Block Adjustment Holding Register Command Sending Data

Field		Bytes	DataType	Remark
Start Register address	addr	2×1	UINT	
Register Number	cnt	2×1	UINT	<=120
Length of Value	len	1	BYTE	len = 2×cnt
Value	val	2×cnt	UINT	

Response:
Table 4-4-2-2 Single regulation holding register command response data

Field		Bytes	DataType	Remark
Start Register address	addr	2×1	UINT	Same as Request
Register Number	cnt	2×1	UINT	Same as Request

4.4. 异常应答机制/ Exception response mechanism

异常应答时，功能码=接收到的功能码+80H，数据为一个字节的异常码。

异常码为对应的异常原因如表 4-6-1:

In case of exception response, function code=received function code+80H, and data is one byte exception code.

The exception code is the corresponding cause of the exception, as shown in Table 4-6-1:

表 4-6-1 异常码对应含义/ Table 4-6-1 Meaning of exception code

异常码/Code	名称/Name	原因/Reason
1	非法功能/ Illegal function	请求命令中的功能码是不允许的操作。可能是从设备在错误的状态中执行请求。 The function code in the request command is not allowed. The slave device may execute the request in the wrong state.
2	非法数据地址/ Illegal address	请求数据中包含的数据地址是不允许的地址。特别当请求命令数据中有起始地址和寄存器数量组合的时候。对于带有 100 个寄存器的从设备来说，带有偏移量 96 和长度 4 的请求会成功，带有偏移量 96 和长度 5 的请求将产生异常码 02。 The data address contained in the request data is not allowed. Especially when there is a combination of starting address and number of registers in the request command data. For the slave device with 100 registers, the request with offset 96 and length 4 will succeed, and the request with offset 96 and length 5 will generate exception code 02.
3	非法数据值/ Illegal Value	请求数据中的数值为不允许的值。可能是数据的长度不正确，也可能是数据的值超过了特定值的有效范围。 The value in the request data is not allowed. The length of the data may be incorrect, or the value of the data may exceed the valid range of a specific value.

异常判断流程参考图 4.6.1 / Refer to Figure 4.6.1 for abnormal judgment flow

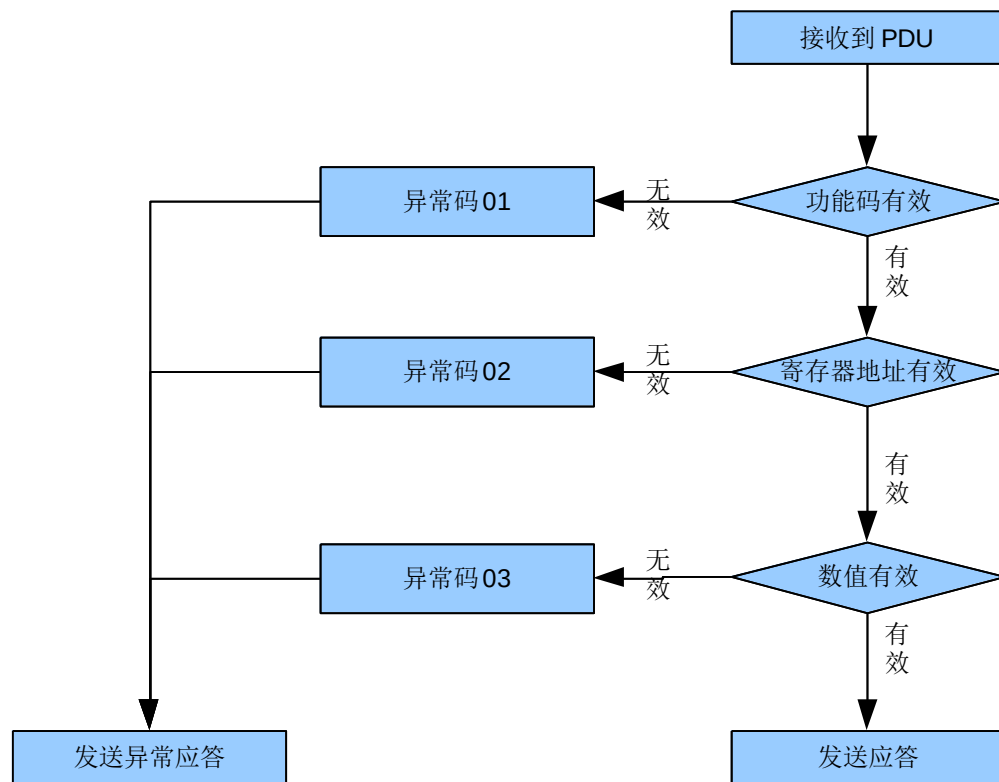


图 4.6.1 异常判断流程

4.5. 常用事务参考定义/ Common Transaction Reference Definitions

结合 ESMU 监控的日常经验，提供以下常用参考事务定义以及相应的“事物处理标识符”。本节不具有强制约束力，实际用户可以根据自己需要定制事务含义。

Combined with the daily experience of ESMU monitoring, the following common reference transaction definitions and corresponding "transaction processing identifiers" are provided. This section is not binding, and actual users can customize the meaning of transactions according to their own needs

4.5.1. ESMU 设备对象常见事务/ Common transactions of ESMU device objects

表 4-5-1 ESMU 设备对象常见事务表/ Table 4-5-1 Common Transactions of ESMU Equipment Objects

事务名 /Name	事务处理标识符/ Code	说明
读取离散/Read Discrete	1	
读取输入/Read Input	2	
读取参数/Read parameters	3	
修改参数/modify parameters	4	

说明：不是所有这些事务都会用到，且事务的具体内容可以有所变化。

4.5.2. ESBCM 对象常见事务/ Common transactions of ESBCM device objects

表 4-7-2 ESBCM 对象常见事务表

事务名 /Name	事务处理标识符/Code	说明
读取参数/Read parameters	0	
读取汇总告警/Read summary alarm	1	
读取单体告警/Read individual alarm	2	
读取汇总输入寄存器/Read summary input register	3	
读取单体电压 1~400 信息/Read unit voltage 1~400 information	10	

说明：事务标识符可以自行定义。上表仅供参考。Note: The transaction identifier can be defined by yourself. The above table is for reference only.

5. 寄存器分布 /Register distribution

5.1. ESMU 设备对象寄存器分布/ESMU Regisiter distribution

5.1.1. 离散输入寄存器表/ Discrete Input Register Table

名称/Name	起始地址（十进制）/ Start address (decimal)	总长度/ Total length
堆数据信息/System data information	1	80
簇 1 数据信息/String 1 data informaton	200	88
簇 2 数据信息/String 2 data informaton	300	88
簇 3 数据信息/String 3 data informaton	400	88



簇 4 数据信息/String 4 data information	500	88
簇 5 数据信息/String 5 data information	600	88
簇 6 数据信息/String 6 data information	700	88
簇 7 数据信息/String 7 data information	800	88
簇 8 数据信息/String 8 data information	900	88
簇 9 数据信息/String 9 data information	1000	88
簇 10 数据信息/String 10 data information	1100	88
簇 11 数据信息/String 11 data information	1200	88
簇 12 数据信息/String 12 data information	1300	88
簇 13 数据信息/String 13 data information	1400	88
簇 14 数据信息/String 14 data information	1500	88
簇 15 数据信息/String 15 data information	1600	88
簇 16 数据信息/String 16 data information	1700	88
簇 17 数据信息/String 17 data information	1800	88
簇 18 数据信息/String 18 data information	1900	88
簇 19 数据信息/String 19 data information	2000	88
簇 20 数据信息/String 20 data information	2100	88

字段/Field	起始地址 (十进制) /	说明/Remark
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	Start address (decimal)	
堆信息具体明细：可以根据项目情况召唤对应的数据，不一定需要全部召唤。 Specific details of heap information: corresponding data can be called according to project conditions, not necessarily all.		
堆内各组端电压欠压轻度告警汇总 Summary of Total voltage undervoltage slight alarm at each String	1	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端电压欠压中度告警汇总 Summary of Total voltage undervoltage Medium alarm at each String	2	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端电压欠压重度告警汇总 Summary of Total voltage undervoltage serious alarm at each String	3	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端电压过压轻度告警汇总 Summary of Total voltage overvoltage slight alarm at each String	4	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端过压过压中度告警汇总 Summary of Total voltage overvoltage Medium alarm at each String	5	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端过压过压重度告警汇总 Summary of Total voltage overvoltage serious alarm at each String	6	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端过流轻度告警汇总 Summary of overcurrent slight alarm at each String	7	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组端过流中度告警汇总	8	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



Summary of overcurrent Medium alarm at each String		
堆内各组端过流重度告警汇总 Summary of overcurrent serious alarm at each String	9	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组绝缘电阻低轻度告警汇总 Summary of Low insulation resistance slight alarm at each String	10	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组绝缘电阻低中度告警汇总 Summary of Low insulation resistance Medium alarm at each String	11	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组绝缘电阻低重度告警汇总 Summary of Low insulation resistance serious alarm at each String	12	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组模块欠温度轻度告警汇总 Summary of Module Low temperature slight alarm at each String	13	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组模块欠温度中度告警汇总 Summary of Module Low temperature Medium alarm at each String	14	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组模块欠温度重度告警汇总 Summary of Module Low temperature serious alarm at each String	15	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组模块过温度轻度告警汇总 Summary of Module Over temperature slight alarm at each String	16	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组模块过温度中度告警汇总 Summary of Module Over temperature Medium alarm at each String	17	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



堆内各组模块过温度重度告警汇总 Summary of Module Over temperature serious alarm at each String	18	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过压轻微告警汇总 Summary of Cell Over voltage slight alarm at each String	19	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过压中度告警汇总 Summary of Cell Over voltage Medium alarm at each String	20	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过压严重告警汇总 Summary of Cell Over voltage serious alarm at each String	21	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠压轻微告警汇总 Summary of Cell Low voltage slight alarm at each String	22	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠压中度告警汇总 Summary of Cell Low voltage Medium alarm at each String	23	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠压严重告警汇总 Summary of Cell Low voltage serious alarm at each String	24	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体压差轻微告警汇总 Summary of Cell differential voltage slight alarm at each String	25	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体压差中度告警汇总 Summary of Cell differential voltage Medium alarm at each String	26	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



堆内各单体压差严重告警汇总 Summary of Cell differential voltage serious alarm at each String	27	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠温轻微告警汇总 Summary of Cell low temperature slight alarm at each String	28	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠温中度告警汇总 Summary of Cell low temperature Medium alarm at each String	29	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体欠温严重告警汇总 Summary of Cell low temperature serious alarm at each String	30	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过温轻微告警汇总 Summary of Cell over temperature slight alarm at each String	31	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过温中度告警汇总 Summary of Cell over temperature Medium alarm at each String	32	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体过温严重告警汇总 Summary of Cell over temperature serious alarm at each String	33	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体温差轻微告警汇总 Summary of Cell differential temperature slight alarm at each String	34	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体温差中度告警汇总 Summary of Cell differential temperature Medium alarm at each String	35	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



堆内各单体温差严重告警汇总 Summary of Cell differential temperature serious alarm at each String	36	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 低轻微告警汇总 Summary of Cell SOC Low slight alarm at each String	37	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 低中度告警汇总 Summary of Cell SOC Low Medium alarm at each String	38	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 低严重告警汇总 Summary of Cell SOC Low serious alarm at each String	39	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 高轻微告警汇总 Summary of Cell SOC High slight alarm at each String	40	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 高中度告警汇总 Summary of Cell SOC High Medium alarm at each String	41	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOC 高严重告警汇总 Summary of Cell SOC High serious alarm at each String	42	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 低轻微告警汇总 Summary of Cell SOH Low slight alarm at each String	43	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 低中度告警汇总 Summary of Cell SOH Low Medium alarm at each String	44	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 低严重告警汇总 Summary of Cell SOH Low serious alarm at each String	45	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 高轻微告警汇总 Summary of Cell SOH High slight alarm at each String	46	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 高中度告警汇总 Summary of Cell SOH High Medium alarm at each String	47	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各单体 SOH 高严重告警汇总 Summary of Cell SOH High serious alarm at each String	48	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



堆内各主控失联汇总/ESBCM Lost Communication	49	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各从控失联告警汇总/ESBMM Lost Communicaion	50	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各组电压异常 (组间压差大于 20V) Abnormal voltage of each string in the System (the total Voltage difference between Strings is greater than 20V)	51	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内接触器断开异常/Abnormal disconnection of contactor	52	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内接触器闭合异常/Abnormal closing of contactor	53	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
充电禁止/ Charging prohibited	54	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
放电禁止/ Discharging prohibited	55	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
BMS 系统告警汇总/ BMS system alarm summary	56	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
BMS 系统故障汇总/ Summary of BMS system faults	57	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
输入 IN0 (根据实际项目空调、急停或者其他硬件信号) Input IN0 (according to the actual project air conditioning, emergency stop or other hardware signals)	58	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
输入 IN1 (根据实际项目空调、急停或者其他硬件信号) Input IN1 (according to the actual project air conditioning, emergency stop or other hardware signals)	59	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
输入 IN2 (根据实际项目空调、急停或者其他硬件信号) Input IN1 (according to the actual project air conditioning, emergency stop or other hardware signals)	60	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
输入 IN3 (根据实际项目空调、急停或者其他硬件信号) Input IN3 (according to the actual project air	61	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



conditioning, emergency stop or other hardware signals)		
预留/reserved	62	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	63	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	64	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	65	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	66	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	67	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	68	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
预留/reserved	69	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各端子过温轻微告警汇总 Summary of terminal over temperature slight alarm at each String	70	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各端子过温中度告警汇总 Summary of terminal over temperature Medium alarm at each String	71	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内各端子过温重度告警汇总 Summary of terminal over temperature serious alarm at each String	72	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内模块电压过压轻微告警汇总 Summary of pack voltage overvoltage slight alarm at each String	73	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内模块电压过压中度告警汇总 Summary of pack voltage overvoltage Medium alarm at each String	74	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内模块电压过压严重告警汇总 Summary of pack voltage overvoltage serious alarm at	75	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



each String		
堆内模块电压欠压轻微告警汇总 Summary of pack voltage undervoltage slight alarm at each String	76	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内模块电压欠压中度告警汇总 Summary of pack voltage undervoltage Medium alarm at each String	77	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
堆内模块电压欠压严重告警汇总 Summary of pack voltage undervoltage serious alarm at each String	78	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
电压采集失联/ Cell Voltage acquisition fault	79	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
温度采集失联/Cell Temperature acquisition fault	80	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇信息具体明细: 可以根据项目情况召唤对应的数据, 不一定需要全部召唤。 Specific details of String information: corresponding data can be called according to project conditions, not necessarily all.		
簇 N 主控通讯失联 / String N ESBCM Lost communication	200	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电压欠压轻度告警 String N Total voltage undervoltage slight alarm	201	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电压欠压中度告警	202	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



String N Total voltage undervoltage Medium alarm		
簇 N 端电压欠压重度告警 String N Total voltage undervoltage serious alarm	203	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电压过压轻度告警 String N Total voltage overvoltage slight alarm	204	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电压过压中度告警 String N Total voltage overvoltage Medium alarm	205	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电压过压重度告警 String N Total voltage overvoltage serious alarm	206	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电流轻度告警 String N over Current slight alarm	207	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电流中度告警 String N over Current Medium alarm	208	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端电流重度告警 String N over Current serious alarm	209	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体欠压轻度告警 String N cell voltage undervoltage slight alarm	210	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体欠压中度告警 String N cell voltage undervoltage Medium alarm	211	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体欠压重度告警 String N cell voltage undervoltage serious alarm	212	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过压轻度告警 String N cell voltage overvoltage slight alarm	213	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过压中度告警 String N cell voltage overvoltage Medium alarm	214	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过压重度告警 String N cell voltage overvoltage serious alarm	215	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



簇 N 单体欠温轻度告警 String N cell low temperature slight alarm	216	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体欠温中度告警 String N cell low temperature Medium alarm	217	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体欠温重度告警 String N cell low temperature serious alarm	218	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过温轻度告警 String N cell over temperature slight alarm	219	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过温中度告警 String N cell over temperature Medium alarm	220	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体过温重度告警 String N cell over temperature serious alarm	221	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 低轻度告警 String N cell SOC low slight alarm	222	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 低中度告警 String N cell SOC low Medium alarm	223	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 低重度告警 String N cell SOC low serious alarm	224	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 高轻度告警 String N cell SOC High slight alarm	225	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 高中度告警 String N cell SOC High Medium alarm	226	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOC 高重度告警 String N cell SOC High serious alarm	227	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOH 低轻度告警 String N cell SOH Low slight alarm	228	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体 SOH 低中度告警	229	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm



String N cell SOC Low Medium alarm		
簇 N 单体 SOH 低重度告警 String N cell SOC Low serious alarm	230	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体压差轻度告警 String N cell voltage different slight alarm	231	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体压差中度告警 String N cell voltage different Medium alarm	232	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体压差重度告警 String N cell voltage different serious alarm	233	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体温差轻度告警 String N cell temperature different slight alarm	234	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体温差中度告警 String N cell temperature different Medium alarm	235	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 单体温差重度告警 String N cell temperature different serious alarm	236	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 从控 1 通讯失联/String N ESBMM#1 lost Communication	237	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
...	...	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 从控 40 通讯失联/String N ESBMM#40 lost Communication	276	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端子温度过高轻度告警 String N Terminal temperature high slight alarm	277	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端子温度过高中度告警 String N Terminal temperature high Medium alarm	278	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 端子温度过高重度告警 String N Terminal temperature high serious alarm	279	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 模块电压过高轻度告警 String N pack voltage overvoltage slight alarm	280	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm

簇 N 模块电压过高中度告警 String N pack voltage overvoltage Medium alarm	281	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 模块电压过高重度告警 String N pack voltage overvoltage serious alarm	282	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 模块电压过低轻度告警 String N pack voltage undervoltage slight alarm	283	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 模块电压过低中度告警 String N pack voltage undervoltage slight alarm	284	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 模块电压过低重度告警 String N pack voltage undervoltage slight alarm	285	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 电压采集故障/String N cell voltage Acquisition fault	286	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm
簇 N 温度采集故障/String N Cell temperature Acquisition fault	287	0--无告警, 1--告警/0 -- No alarm, 1 -- Alarm

5.1.2. 保持寄存器表/ Holding register table

字段/Field	起始地址（十进制） Start Address (Decimal)	数据类型 Type	说明/Rmark
电池组数/String Number	500	UINT	1~20
系统故障复位/ System fault reset	501	UINT	0: 不复归 1: 复归 其它值: 无效 0: Do not reset 1: Reset other values: Invalid
电操控制/ Main circuit breaker control	502	UINT	0: 不操作 1: 合闸 2: 分闸 其它值: 无效 0: No operation 1: Closing 2: Opening Other values: Invalid



系统上下电控制指令 System power up and down control command	503	UINT	注：默认不使用，供电后堆自主上电，若需要，请于项目对接阶段明确 0：不操作 1：上高压 2：下高压 其它值：无效 Note: It is not used by default, and the reactor is powered on automatically after power supply. If necessary, please specify in the project docking stage 0: No operation 1: Power on 2: Power off Other values: Invalid
簇 N 维护模式控制 String N maintenance mode control	504~523	UINT	N 范围：1~20 --- 同一时刻，只允许对一簇进行维护； 0：无效 1：上电 2：下电 N range: 1~20 --- only one cluster can be maintained at the same time; 0: invalid 1: power on 2: power off
RTC-年/RTC-Year	524	UINT	分辨率：1 偏移量：2000 Resolution: 1 Offset: 2000 数据范围：0~100 data field:0~100
RTC-月/RTC-Month	525	UINT	数据范围：1~12/ data field:1~12
RTC-日/RTC-Day	526	UINT	数据范围：1~31/ data field:1~31
RTC-时/RTC-Hour	527	UINT	数据范围：0~23/ data field:0~23
RTC-分/RTC-Minute	528	UINT	数据范围：0~59/ data field:0~59



RTC-秒/RTC-Second	529	UINT	数据范围: 0~59/ data field:0~59
心跳/HeartBeat	530	UINT	数据范围: 0~65535/ data field:0~65535
项目外设信息/ Project peripheral information	1000~2000		项目外设信息添加至 1000~2000 的地址内/ Item peripheral information is added to addresses from 1000 to 2000

5.1.3. 输入寄存器表/ Input register table

名称/Name	起始地址（十进制）/Start Address (decimal)	总长度/Total Length
堆数据信息/System data	1	47
簇 1 数据信息/String 1 data	100	2991
簇 2 数据信息/String 2 data	3100	2991
簇 3 数据信息/String 3 data	6100	2991
簇 4 数据信息/String 4 data	9100	2991
簇 5 数据信息/String 5 data	12100	2991
簇 6 数据信息/String 6 data	15100	2991
簇 7 数据信息/String 7 data	18100	2991



簇 8 数据信息/String 8 data	21100	2991
簇 9 数据信息/String 9 data	24100	2991
簇 10 数据信息/String 10 data	27100	2991
簇 11 数据信息/String 11 data	30100	2991
簇 12 数据信息/String 12 data	33100	2991
簇 13 数据信息/String 13 data	36100	2991
簇 14 数据信息/String 14 data	39100	2991
簇 15 数据信息/String 15 data	42100	2991
簇 16 数据信息/String 16 data	45100	2991
簇 17 数据信息/String 17 data	48100	2991
簇 18 数据信息/String 18 data	51100	2991
簇 19 数据信息/String 19 data	54100	2991
簇 20 数据信息/String 20 data	57100	2991

字段/Field	起始地址 (十进制) Start Address (Decimal)	数据类型/Type	说明/Remark
堆信息具体明细：可以根据项目情况召唤对应的数据，不一定需要全部召唤。 Specific details of String information: corresponding data can be called according to project conditions, not necessarily all.			
电池堆电操状态/System Circuit breaker status	1	UINT	1:合 0:分 (需要项目有电操反馈节点) 1: Closing 0: opening (the project needs electric operation feedback node)
电池堆电压/ System Total Voltage	2	UINT	0.1V/bit
电池堆电流/ System Current	3	UINT	0.1A/bit 偏移量/Offset: -1600.0A



电池堆 SOC/ System SOC	4	UINT	1%/bit
电池堆 SOH/ System SOH	5	UINT	1%/bit
最高电池电压/ Maximum battery voltage	6	UINT	0.001V/bit
最高电压电池组号 Maximum voltage battery Sring No	7	UINT	1/bit
最高电压电池所在组中的点号 Point number in Maximum voltage battery Sring	8	UINT	1/bit
最低电池电压 Minimum battery voltage	9	UINT	0.001V/bit
最低电压电池组号 Minimum voltage battery Sring No	10	UINT	1/bit
最低电压电池所在组中的点号 Point number in Minimum voltage battery Sring	11	UINT	1/bit
最高电池温度 Maximum battery temperature	12	UINT	1℃/bit , 偏移/Offset: -40℃
最高温度电池组号 Maximum temperature battery Sring No	13	UINT	1/bit
最高温度电池所在组中的点号 Point number in Maximum temperature battery Sring	14	UINT	1/bit
最低电池温度 Minimum battery temperature	15	UINT	1℃/bit, 偏移/Offset: -40℃
最低温度电池组号	16	UINT	1/bit



Minimum temperature battery String No			
最低温度电池所在组中的点号 Point number in Minimum temperature battery String	17	UINT	1/bit
堆累计充电电量 System Accumulated charging capacity	18	UINT32	0.1KWh/bit
堆累计放电电量 System Accumulated discharging capacity	20	UINT32	0.1KWh/bit
堆单次累计充电电量 System Accumulated charge quantity for a single time	22	UINT32	0.1KWh/bit
堆单次累计放电电量 System Accumulated discharge quantity for a single time	24	UINT32	0.1KWh/bit
堆可充电量 System Rechargeable capacity	26	UINT32	0.1KWh/bit
堆可放电量 System Discharge capacity	28	UINT32	0.1KWh/bit
可用放电时间 Available discharge time	30	UINT	分钟/bit Min/bit
可用充电时间 Available charging time	31	UINT	分钟/bit Min/bit
允许最大放电功率 Allowable maximum discharge	32	UINT	0.1KW/bit



power			
允许最大充电功率 Allowable maximum charge power	33	UINT	0.1KW/bit
允许最大放电电流 Allowable maximum discharge current	34	UINT	0.1A/bit 偏移量/offset: -1600.0A
允许最大充电电流 Allowable maximum charge current	35	UINT	0.1A/bit 偏移量/offset: -1600.0A
当天放电次数 Discharge times of the day	36	UINT	1 次/bit
当天充电次数 charge times of the day	37	UINT	1 次/bit
当天放电电量 Discharge quantity of the day	38	UINT32	0.1KWh/bit
当天充电电量 Charge quantity of the day	40	UINT32	0.1KWh/bit
运行温度 Operating temperature	42	UINT	1℃/bit , 偏移/offset: -40℃, 电池堆平均温度
BMS 堆当前状态/ System State	43	UINT	0x00: 初始状态 0x01: 充电 0x02: 放电 0x03: 就绪 0x04: 簇维护 0x05: 禁充 0x06: 禁放 0x07: 充放禁止 0x08: 故障 0x09: 故障恢复 0x0A: 测试模式 0x0B: 下电中 0x0C: 下电完成 其它值: 预留 0x00: initial state 0x01: charging 0x02: discharging 0x03: ready 0x04: cluster maintenance 0x05: charge prohibition 0x06: discharge prohibition 0x07: Charging and discharging prohibited 0x08: Fault 0x09: Fault recovery 0x0A: test mode 0x0B: power-off 0x0C:



			Power-off complete Other values: reserved 补充描述/ supplemental description: 初始化——系统上电自检期间，继电器未闭合/ Initialization - The relay is not closed during the system power-on self-test 充电——继电器闭合后，系统检测到所有簇均有充电电流/ Charging - After the relay is closed, the system detects charging current in all clusters 放电——继电器闭合后，系统检测到所有簇均有放电电流/ Discharge -- After the relay is closed, the system detects discharge current in all clusters 就绪——继电器闭合后，无充电、无放电情况/ Ready: After the relay is closed, there is no charging or discharging 簇维护——人为点击显控簇维护按钮进形维护功能/ Cluster maintenance: Manually click the display control cluster maintenance button to enter the shape maintenance function 禁充——继电器闭合，系统禁止充电，允许放电/ No charge -- The relay is closed, the system does not charge, allow discharge / 禁放——继电器闭合，系统允许充电，禁止放电/ Discharges: - The relay is closed and the system allows charging. Discharges are prohibited 充放禁止——继电器闭合，系统禁止充电，禁止放电/ Charging and discharging prohibition: The relay is closed, and the system cannot charge or discharge 故障——系统严重故障，需要下高压的告警/ Fault -- The system is seriously faulty and requires a high voltage alarm
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			故障恢复---告警下高压后, 电池系统严重故障解除/ Fault recovery -- After the high voltage alarm is generated, the battery system is cleared of serious faults 测试模式---使用我司上位机强制控制过主控 D0 功能后, 进行强行测试模式, 此时无常规电池保护, 需谨慎/ Test mode -- After using the upper computer of our company to forcibly control the main D0 function, we will carry out the forced test mode without conventional battery protection, so it is necessary to be cautious 下电中: 电池系统无故障, 并且控制权限由外设启用时, 收到下电指令后执行中的状态/ Power-off: When the battery system is not faulty and the control permission is enabled by the peripheral, the battery system is executed after receiving the power-off command 下电完成: 电池系统无故障, 并且控制权限由外设启用时, 收到下电指令后执行完毕的状态/ Power-off completed: When the battery system is normal and the control permission is enabled by the peripheral, the power-off is completed after receiving the power-off command
BMS 充放电状态/Charge discharge state	44	UINT	0x00: 其它 0x01:放电 0x02:充电 0x00: Others 0x01: Discharge 0x02: Charge
电池堆绝缘电阻/ System Insulation resistance	45	UINT	1K Ω /bit
PCS 和 BMS 通信故障/ PCS and BMS communication failure	46	UINT	0x01:通讯失联 0x00:通讯正常 0x01: communication loss 0x00: communication is normal
EMS 和 BMS 通信故障/ EMS and BMS communication failure	47	UINT	0x01:通讯失联 0x00:通讯正常 0x01: communication loss 0x00: communication is normal



堆累计充电时间/ Pile cumulative charging time	48	UINT32	单位：秒，分辨率：1/ Unit: second, resolution: 1
堆累计放电时间/ Pile cumulative discharging time	50	UINT32	单位：秒，分辨率：1/ Unit: second, resolution: 1
簇信息具体明细：可以根据项目情况召唤对应的数据，不一定需要全部召唤。			
簇 N 电池簇状态/String N State	100	UINT	0x00：初始状态 0x01：充电 0x02：放电 0x03：就绪 0x04：簇维护 0x05：禁充 0x06：禁放 0x07：充放禁止 0x08：故障 0x09：故障恢复 0x0A：测试模式 其它值：预留 0x00: initial state 0x01: charging 0x02: discharging 0x03: ready 0x04: cluster maintenance 0x05: charge prohibition 0x06: discharge prohibition 0x07: Charging and discharging prohibited 0x08: Fault 0x09: Fault recovery 0x0A: test mode Other values: reserved
簇 N 允许充电最大功率 String N Maximum allowable charging power	101	UINT	分辨率：0.1 单位：KW Resolution: 0.1 Unit: KW
簇 N 允许放电最大功率 String N Maximum allowable discharging power	102	UINT	分辨率：0.1 单位：KW Resolution: 0.1 Unit: KW
簇 N 允许充电最大电压 String N Maximum allowable charging Voltage	103	UINT	分辨率：0.1 单位：V Resolution: 0.1 Unit: V



簇 N 允许放电最大电压 String N Maximum allowable discharging Voltage	104	UINT	分辨率: 0.1 单位: V Resolution: 0.1 Unit: V
簇 N 允许充电最大电流 String N Maximum allowable charging Current	105	UINT	分辨率: 0.1 单位: A Resolution: 0.1 Unit: A
簇 N 允许放电最大电流 String N Maximum allowable discharging Current	106	UINT	分辨率: 0.1 单位: A Resolution: 0.1 Unit: A
簇 NDI1/String N Input1	107	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI2/String N Input2	108	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI3/String N Input3	109	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI4/String N Input4	110	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI5/String N Input5	111	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI6/String N Input6	112	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI7/String N Input7	113	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 NDI8/String N Input8	114	UINT	1: 信号输入 0: 信号未输入 1: Signal input 0: signal not input
簇 N 组电压/String N total Voltage	115	UINT	0.1V/bit
簇 N 组电流/String N Current	116	UINT	0.1A/bit, 偏移量/offset: -1600.0A
簇 N 模块温度/Sting N's ESBCM module temperature	117	UINT	1°C/bit , 偏移量/offset: -40°C
簇 N 组 SOC/ String N SOC	118	UINT	1%/bit
簇 N 组 SOH/String N SOH	119	UINT	1%/bit
簇 N 组绝缘电阻/String N insulation resistance	120	UINT	1KΩ/bit
簇 N 平均单体电压/String N Average cell voltage	121	UINT	0.001V/bit



簇 N 平均单体温度//String N Average cell temperature	122	UINT	1℃/bit, 偏移/offset: -40℃
簇 N 最高单体电压 /String N Maximum cell voltage	123	UINT	0.001V/bit
簇 N 最高单体电压对应点号/Cell Number of String N Maximum cell voltage	124	UINT	1/bit
簇 N 最低单体电压 /String N Minimum cell voltage	125	UINT	0.001V/bit
簇 N 最低单体电压对应点号/Number of String N Minimum cell voltage	126	UINT	1/bit
簇 N 最高单体温度 String N Maximum cell temperature	127	UINT	1℃/bit, 偏移/offset: -40℃
簇 N 最高单体温度对应点号 /Number of String N Maximum cell temperature	128	UINT	1/bit
簇 N 最低单体温度 /String N Minimum cell temperature	129	UINT	1℃/bit, 偏移/offset: -40℃
簇 N 最低单体温度对应点号/Number of String N Minimum cell temperature	130	UINT	1/bit
簇 N 最高单体 SOC/ String N Maximum cell SOC	131	UINT	1%/bit
簇 N 最高单体 SOC 对应点号/ Number of String N Maximum cell SOC	132	UINT	1/bit
簇 N 最低单体 SOC / String N Minimum cell SOC	133	UINT	1%/bit



簇 N 最低单体 SOC 对应点号 /Number of String N Minimum cell SOC	134	UINT	1/bit
簇 N 最高单体 SOH / String N Maximum cell SOH	135	UINT	1%/bit
簇 N 最高单体 SOH 对应点号 /Number of String N Maximum cell SOH	136	UINT	1/bit
簇 N 最低单体 SOH/ String N Minimum cell SOH	137	UINT	1%/bit
簇 N 最低单体 SOH 对应点号 /Number of String N Minimum cell SOH	138	UINT	1/bit
簇 N 累计充电电量 /String N Accumulated charging capacity	139	UINT32	0.1KWh/bit
簇 N 累计放电电量 /String N Accumulated discharging capacity	141	UINT32	0.1KWh/bit
簇 N 单次累计充电电量/String N Accumulated charge quantity for a single time	143	UINT32	0.1KWh/bit
簇 N 单次累计放电电量 /String Accumulated discharge quantity for a single time	145	UINT32	0.1KWh/bit
簇 N 可充电量/String N Rechargeable capacity	147	UINT32	0.1KWh/bit
簇 N 可放电量/String N Discharge capacity	149	UINT32	0.1KWh/bit



簇 N 箱 1SOC/ String N pack 1 SOC	151	UINT	1%/bit
.....		UINT	1%/bit
簇 N 箱 40SOC/ String N pack 40 SOC	190	UINT	1%/bit
簇 N 单体电压 001/ String N cell voltage 001	191	UINT	0.001V/bit
.....		UINT	0.001V/bit
簇 N 单体电压 700/ String N cell voltage 700	890	UINT	0.001V/bit
簇 N 单体温度 001 / String N cell temperature 001	891	UINT	1℃/bit, 偏移/offset: -40℃
.....		UINT	1℃/bit, 偏移/offset: -40℃
簇 N 单体温度 700 / String N cell temperature 700	1590	UINT	1℃/bit, 偏移/offset: -40℃
簇 N 单体 SOC 001 / String N cell SOC001	1591	UINT	1%/bit
.....		UINT	1%/bit
簇 N 单体 SOC 700 / String N cell SOC700	2290	UINT	1%/bit
簇 N 单体 SOH 001 / String N cell SOH001	2291	UINT	1%/bit
.....		UINT	1%/bit
簇 N 单体 SOH 700 / String N cell SOH700	2990	UINT	1%/bit
簇 N 端子温度 001 /String N Terminal temperature 001	2991	UINT	1℃/bit, 偏移/offset: -40℃
.....			
簇 N 端子温度 100	3090	UINT	1℃/bit, 偏移/offset: -40℃



杭州高特电子有限公司

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A3 (第49页/共 49 页)

/String N Terminal temperature 100			